IlluminatOR: Learning Compressed Illumination Fields for Object Reshading

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Abstract

We describe IlluminatOR, a method that can convincingly reshade novel objects inserted in an image scene using only image information. IlluminatOR is learned, but uses only in-the-wild images as training data – for example, there are no pairs of images with and without objects. IlluminatOR is trained to compose images from their intrinsics – albedo, shading, and gloss (these come from an image-decomposition model which is pretrained on a synthetically generated dataset). Our architecture means that Illuminator must reconstruct a detailed shading field from an albedo and a summary representation of shading. This reconstruction procedure generalizes to produce a shading field for cut-and-paste albedos that is natural and consistent. We evaluate our method by comparing it against the recent state-of-the-art deep image prior based object reshading method, a state-of-the-art image harmonization baseline. The evaluation shows strong improvements qualitatively, quantitatively, and in overall rendering speed.

1 Introduction

Imagine effortlessly inserting new objects into a scene with a simple drag-and-drop method with realistic lighting and shading effects. Naïve cut-and-paste object insertions produce unrealistic images because the shading of the inserted object does not match the target scene's lighting. A successful image-based insertion system should reshade inserted novel objects such that they are consistent with the target scene's lighting. Further, the system must be able to convincingly insert without requiring a geometric model of the inserted object or the target scene's environment maps. Current object insertion systems rely on the geometry of the object and hence can only insert computer graphics objects and cannot realistically insert real objects with complex surface properties like a furry cat [27, 10]. Moreover, these methods require environment maps or panoramas to reason for the scene's lighting. Furthermore, recovering 3D objects from a single image is still an open and challenging computer vision task [26, 11, 4]. In contrast, in this work, we show a system that convincingly reshades images using only standard datasets like MSCOCO [30] and ADE20k [44] with no ground truth labels.

Our reshading system, IlluminatOR, is entirely an image-based approach and does not require geometry or environment maps. Training such a reshading system with paired (scene-without and scene-with-object) images would be tedious and time-consuming. Instead, we train IlluminatOR to learn a compressed illumination field or CIF by composing images from their intrinsics. We show injecting CIF to a styleGAN architecture will allow it to map albedos to images correctly. When the albedo is a composite with new objects, the imputed shading is convincing. Since we get our intrinsics from a (pretrained) model that uses no labeled data, IlluminatOR is self-supervised.

For our experiments, we show strong qualitative and quantitative convincing object insertions while training IlluminatOR only to compose scenes from their intrinsics. We compare our IlluminatOR insertions with the recently proposed deep image prior based reshading (DIPR) method [3] and an

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Target Scene | Cut-and-Paste | Image Harmonization [5] | DIP Reshading [3] | IlluminatOR (ours)

Figure 1: Image-based object insertion requires convincing object reshading. Current SOTA methods, both image harmonization and DIP reshading fail in complex settings like the example shown here. However, our method – IlluminatOR can convincingly reshade objects only by extrapolating "light style" captured from the target scene. We not only reshade object but also adjust the background realistically as if the inserted objects is native to the target scene. Notice a ring-like reflex, simulating interreflection effects, created on the floor below the car deep inside the scene.

image harmonization [5] baseline. Furthermore, unlike DIPR which takes several minutes to insert an object we show IlluminatOR can realistically insert them in real-time even without trained for object insertions. We also show IlluminatOR will sometimes add fairly reasonable cast-shadows onto the background of the seen which the DIP reshading can not. We also provide empirical evidence that vision transformers are better discriminators for our task possibly because of the patch-level and global level fused feature learning in them.

2 Related Work

Computer Graphics (CG) Object Insertion. Karsch et al.[21, 22] convincingly inserts CG objects into inverse rendering models with a very limited user annotated input. However, building such a rendering model only from an image fragment is extremely challenging as one cannot faithfully recover convincing 3D geometry from a single image; even with SOTA methods [26, 11]. Liao et al.[28, 29] showed that a weak geometric model of the object can correct shading *if* one has strong geometric information about the target scene. An alternative is to learn to predict scene lighting [8–10, 37, 38, 27] from their environment maps and then insert CG objects using external software from their geometric and physical appearance models. In comparison, our method neither requires environment maps to train models for lighting prediction nor depends on the geometry and physical appearance models.

Image-based Object Insertion. Poisson blending [34, 18] based object insertions often lead to color bleeding from the background into the inserted object resulting in color changes, strong boundary artifacts, and displeasing results. Lalonde et al. [23] extend object insertion to a data-driven approach by retrieving objects that match target scene lighting. However, they cannot reliably insert objects if their lighting does not match that of the target scene. Recently, semantic context-based editing was introduced [1, 25]. These methods require labeled annotations and strongly rely on contexts. Our methods do not require any labeled datasets and hence can insert any given object to any given target scene. Another alternative is to use image harmonization (IH) methods for reshading. IH methods are trained to correct corrupted images where a fragment has been adjusted by a color editing process to the original image [41, 42, 5]. In contrast, for object insertions, one must only adjust the shading appearance of the object without altering their colors. We compare our method against [5] (SOTA IH method) and refer to this method as *IH* in our paper.

Recent single-image relighting methods relight only portrait faces under directional lighting [39, 45, 32] and cannot relight/reshade complex objects. Reshading is also quite similar to relighting but has a very notable difference. For reshading, only a fragment or portion of the image has been

reshaded while in relighting, the entire scene is reshaded to a given target lighting condition. We wish to only reshade novel objects but also show IlluminatOR has potential advantages for extensions to image-based relighting of general scenes.

More recently, [3] show leveraging deep image prior (DIP) [43] with consistent image decomposition inferences can reshade cut-and-paste images convincingly. Since their method relies on a DIP, it takes several minutes for them to render a reshaded image. In contrast, our method is just a single pass feed-forward approach to render reshaded images. Furthermore, they rely on pretrained surface normals to learn an explicit lighting representation of the scene. In comparison, we show we can implicitly learn strong lighting representation of the scene directly from image priors and without requiring any surface normals. Finally, since our method is not single-image based we can reshade several objects and evaluate our results for realism using FID based metrics. We use [3] as our strong baseline and refer to this method as *DIPR* in our paper.

3 Object Reshading

Object reshading was first introduced in [3]. The goal is to synthesize a reshaded object transferred from the source image into a target scene image using a simple cut-and-paste operator. They use a deep image prior (DIP) [43] as a neural renderer for reshading. They train DIP to produce consistent image decomposition inferences. A DIP must produce an image such that the albedo of the rendered image should match with the albedo of the cut-and-paste image; shading and gloss outside the inserted object for the reshaded image should match the target scene's shading and gloss fields. Finally, they require the shading from the rendered image to be consistent with the cut-and-paste normals. While the method produces plausible reshading of objects but is heavily reliant on a DIP that operates on a single image at a time with several minutes to render them. Though the method produces reasonable reshading, we would like to build a system that is significantly faster and can reshade objects in real-time. Further, they use surface normals and use an explicit lighting loss (spherical harmonics). Therefore, their reshading quality depends on the quality of the normals. In contrast, we take an alternate approach and learn to form an implicit lighting representation of the scene, which we can then reuse to reshade new objects added to the scene.

Similar to [3], we hypothesize that the overall illumination of the scene does not change when we add new objects to it. Hence, one can learn a compressed illumination field (CIF) of the original target scene, which can then render images with new objects using the same CIF. Therefore, we train our model, not to reshade objects but only to compose images from their albedo, shading, and gloss fields. By doing so, we can implicitly form a strong CIF for images which we can then repurpose for object reshading. To synthesize reshading of the inserted novel objects to the scene, we only add objects to their albedo as they are independent of the lighting of the scene and use the inferred CIF from the target scene to compose an image from the trained image composition model. This results in convincing object reshading as an outcome; supporting our claim that the method can learn a strong summary lighting representation of the scene. We summarize our method in Fig. 2.

4 Learning Compressed Illumination Fields (CIF)

Intrinsic Image Decomposition. Our method is built upon learning a compact lighting representation by composing images from their image intrinsics. As we show in Fig. 2, we train a model that learns to form an image from its albedo, shading and gloss. We obtain these image intrinsics from a pretrained image-decomposition model. One could use any reliable image decomposition model for generating image intrinsis. But as noted in [3], for image reshading it is crucial to preserve high-frequency and fine-grained details in albedo and not in shading. Therefore, we use the same image decomposition model that we obtained from [3] for IlluminatOR. Their image decomposition model is trained on a synthetically generated dataset built on principles outlined in [24, 7], a Retinex model. Hence the model do not require any labeled annotated image intrinsiscs dataset and they show the model generalizes quite well for real images. We refer readers to the paper for details about their intrinsic image decomposition model.

StyleGAN and SPADE. StyleGAN [20] is a style-based generative adversarial network [12] where they learn a mapping of noise vectors into a representation ("style") by a sequence of multi-layer perceptrons. They then inject style vector into the generative model using adaptive instance normalization (AdaIN) [16] and generate high-quality images. SPADE [33] extends this further to a spatially

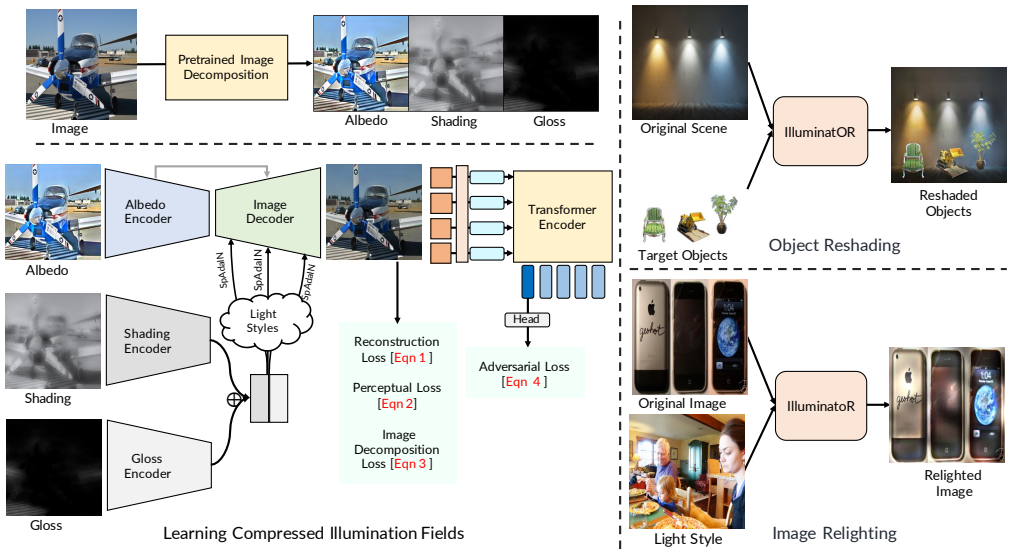


Figure 2: **IlluminatOR overview.** We learn a compressed representation of illumination fields (CIF) of the scene by learning to compose images from their intrinsics – albedo, shading, and gloss fields. We get intrinsics from an off-the-shelf pretrained image decomposition network. We train our network to form an implicit encoding of the illumination of the scene from their shading and gloss fields. We use this illumination encoding as "light styles", and inject them into a U-Net decoder. The U-Net learns to construct images from albedo with spatial conditioning on the injected light styles. We train our network with standard image reconstruction losses and image decomposition consistency loss. Since the pretrained image decomposition network has residuals left behind, we also add adversarial losses using a vision transformer-based discriminator to improve the realism of image compositions. Once trained we show the network learns to form a compact lighting representation of the scene and can convincingly reshade novel objects that are added to them. Interestingly, we also find the CIF to an extent can also relight the entire scene with light styles extracted from random images. For object reshading, we swap original scene's albedo with cut-and-paste albedo. For image relighting, we swap the original scene's shading and gloss with the desired light style's shading and gloss.

adaptive instance normalization (SpAdaIN) by infusing semantic segmentation as spatial-style cues to synthesize images from their segmentation maps. We cannot directly use StyleGAN for reshading as it is hard to disentangle lighting representation. We also do not wish to use SPADE as they limit reproducing images only for object instances for which labeled annotations are available during training. Likewise, CycleGAN [46] require datasets into two different settings which are difficult to get for images with and without objects that are to be added to the scene and cannot help with reshading. However, we take inspiration from StyleGAN and SPADE to design our architecture for learning a compact lighting representation of the scene which can then be used to reshade novel objects when added to it.

Method Overview. Unlike DIPR [3], we do not train our model to reshade images. We train our model only to compose images from intrinsics. We learn CIF and show reshading as a byproduct of learning this CIF. So our objective is to build a learned summary representation of the lighting of the scene. Recall that the illumination of a scene is tied to its shading (+ gloss in our case) and not albedo. We encode a spatial representation from both shading and gloss respectively as "light styles" and concatenate them. We inject these light styles into the decoder using SpAdaIn as shown in Fig 2.

Training Dataset. We could use any dataset. We chose standard datasets available and train our model on a mix of MSCOCO [30] and ADE20K [44]. We use all the training (20k) images from ADE20K but only use randomly chosen 62k images from MSCOCO for training. We first generate a cache of albedo, shading and gloss of all these MSCOCO and ADE20K using the pretrained image decomposition model as described earlier.

Losses. Our losses to train our image composition model encourages composed image to produce that are close to the original image I_O. We use $\mathcal{L}_1$ image reconstruction loss between the original image (I_O) and the composed image from our network (I_C) and also perceptual losses from a VGG feature extractor (Φ) at different feature layers (j) between these images.

$$\mathcal{L}_{recon} = ||I_O - I_C||_1 \tag{1}$$
$$\mathcal{L}_{percp} = ||\Phi_j(I_O) - \Phi_j(I_C)||_2 \tag{2}$$

We also add an image decomposition-based consistency loss as a regularization that helps to compose images with similar image decomposition statistics without deviating while composing images. So we have losses for all three intrinsics from the original image (A_O, S_O, G_O for albedo, shading and gloss resp.), and the composed image (A_C, S_C, G_C) We use $\mathcal{L}_1$ loss for all of them. Therefore, our image decomposition consistency loss:

$$\mathcal{L}_{decomp} = ||A_O - A_C||_1 + ||S_O - S_C||_1 + ||S_O - S_C||_1 \tag{3}$$

ViT Discriminator. The above losses discussed however results in smooth images, do not completely preserve albedo information and produce blurry boundaries. To alleviate this we use adversarial losses. We tried several variants of discriminator – PatchGAN [46], U-Net [36], perceptual discriminator [40] etc., but found unstable training and gradient explosion inspite of using gradient clipping [13], two time-scale update rule (TTUR) [15], label swapping and other tricks[1]. Eventually, we found a vision transformer (ViT) [6] based discriminator to provide stable training with adversarial losses. Our intuition for using them is that ViT has both local (path-wise embedding) and global (self-attention) feature learning mechanism and hence can be a strong discriminator.

ViT is effective in several visual discriminative tasks [14, 35] and scholars also found them to be effective in producing comparable if not the best results for generating images [17, 19]. However, we are not aware of other work that uses a ViT *only* as a discriminator for adversarial training. For our adversarial losses we use non-saturating adversarial loss. Our generator G is the image composition model with parameters ψ and Discriminator D with parameters θ. So our adversarial losses are

$$\mathcal{L}_{\text{adv}} = f(D_\psi(G_\theta(I_C))) + f(-D_\theta(I_O)), \text{where } f(u) \triangleq -\log(1 + \exp(-u)). \tag{4}$$

Our overall losses are

$$\mathcal{L}_{comp} = \lambda_{recon}\mathcal{L}_{recon} + \lambda_{percp}\mathcal{L}_{percp} + \lambda_{decomp}\mathcal{L}_{decomp} + \lambda_{adv}\mathcal{L}_{adv} \tag{5}$$

We train our models for 10 epochs with $\lambda_{recons} = 10$, $\lambda_{percep} = 0.01$, $\lambda_{decomp} = 10$ and $\lambda_{adv} = 0.1$ for the first five epochs and use $\lambda_{recons} = 15$, $\lambda_{percep} = 0.05$, $\lambda_{decomp} = 15$ and $\lambda_{adv} = 0.1$ for the remaining five epochs. These hyperparameters were tuned for stable GAN training. For our experiments we use batch size of 16 on a single Quadro RTX 6000 24GB GPU. Architecture details and ablations for losses are available in supplementary.

To summarize, our network and losses are designed to learn a compact lighting representation of the scene. We train models to map from albedos to images conditioned on the light styles that we obtain from the shading and the gloss fields. Albedos help us to extract color and local changes within images and light styles help in generating images that are consistent with the scene's lighting. Once trained we make use of the learned CIF for object reshading.

5 Experimental Analysis

Once our model has learned to compose images from image intrinsics we directly use our the model – IlluminatOR, for object reshading with no additional training or finetuning or any other supervision. Given a target scene I_t and a source image I_s with objects that are to be transplanted to the target scene with a mask M as locale. We first obtain a cut-and-paste albedo A_{cp} from the pretrained image decomposition model. Next, extract CIF_t from the target image's shading S_t and gloss G_t fields. We then supply this information to IlluminatOR and let it compose an image $I_c = G(A_{cp}; CIF_t)$ as the final reshading image without any post processing steps.

We can render composites using a naïve composition between cut-and-paste albedo with target scene shading and gloss fileds ($I_{naive} = A_{cp} \times S_t + G_t$). We refer this naïve composition as *Naive Comp* in the rest of the paper. In Fig. 3, we first show IlluminatOR can reshade objects even though we did not train it to do so. This shows the method has able to form a less altered lighting representation of the scene from the shading and gloss fields to reshade these objects. For spheres, DIPReshading has better sphere surface reconstruction from 2D disks (because they use normals). Our shadings seem to be better than DIPReshading, likely because IlluminatOR can add colored interreflection effects.

[1] https://github.com/soumith/ganhacks

Figure 3: **Rendering Spheres.** IH produces a mean color of the target scene as reshading. Naive composition leads to a cross-talk between the background and the foreground, resulting in a change of solid disk appearance to translucent. DIPR produces convincing reshading and has a slightly better notion of shape because of their use of surface normals and spherical harmonics. IlluminatOR produces a plausibly realistic shading when trained to only compose images with no geometric or environment or any other external supervision.

5.1 Evaluations

We experimented with 2D disc insertions as spheres and inserted a lego bull-dozer, a ficus, a chair and a set of 16 different reflective materials obtained from Mildenhall et al. [31] (all the images obtained from [3]). We also perform extensive evaluations for bunny insertions and provide a comparison against SOTA insertion methods which require a geometric and physical model of the inserted objects. Further these methods (not [2]) require labeled training data to predict environment maps for rendering using geometric and physical models. We refer to them as supervised (sup.) baselines.

For the bunny insertions, we could only perform a qualitative comparisons with DIPR as DIPR requires about 15 minutes to render a single reshaded image. We have four SOTA methods [2, 8, 10, 27] that renders bunnies into 20 different scenes obtained from Garon et al. [10]. So we use each method's insertions as a source for our bunnies albedo and construct cut-and-paste images. In total we construct 80 cut-and-paste images for 20 different scenes from four different methods. Note these methods do not provide 2D masks for the location of inserted bunnies. We therefore manually extracted masked regions from insertions done with Barron et al.[2]. [2] do not have any cast-shadows and other affects so we only extract masked locations of the bunnies alone. Also, there could be certain boundary and misalignment errors from scaling (different resolutions) and hand annotations.

We reshade 80 images from 20 different scenes spread across four albedo sources. For naive composition (Naive) baseline and IlluminatOR (Ours), we only need albedo of bunnies from these methods. We use shading and gloss fields from the target scene (see Fig. 5) For IH baseline we use the cut-and-paste image directly (which has the entire rendering information) but still the method suffers to reshade. Moreover, to better understand to what extent the method reshades bunnies well, and how they actually compare to the source method we report RMSE and PSNR for the object alone in our Tab.1. Other details in supplementary.

Observations. Naive intrinsic image composition (Naive) do not change any other pixels other than the object itself (suffer mildly due to residuals from decomposition network) However, the object coherence isn't preserved when using naive composition method. Quantitatively, the difference between our reshadings are baseline reshadings become apparent when the quality of the albedo from source improves significantly. We get strong reshading performance from IlluminatOR when we have less corrupted and convincing albedo fields from the source. However, in our qualitative figure we still report renderings from Gardner et al [8] as source for our albedo.

6 Discussion and Conclusion

Extending IlluminatOR to scene relighting. As a stress test for our learned CIF representations, we also experimented to see if IlluminatoR can relight the entire scene given a new lighting style from

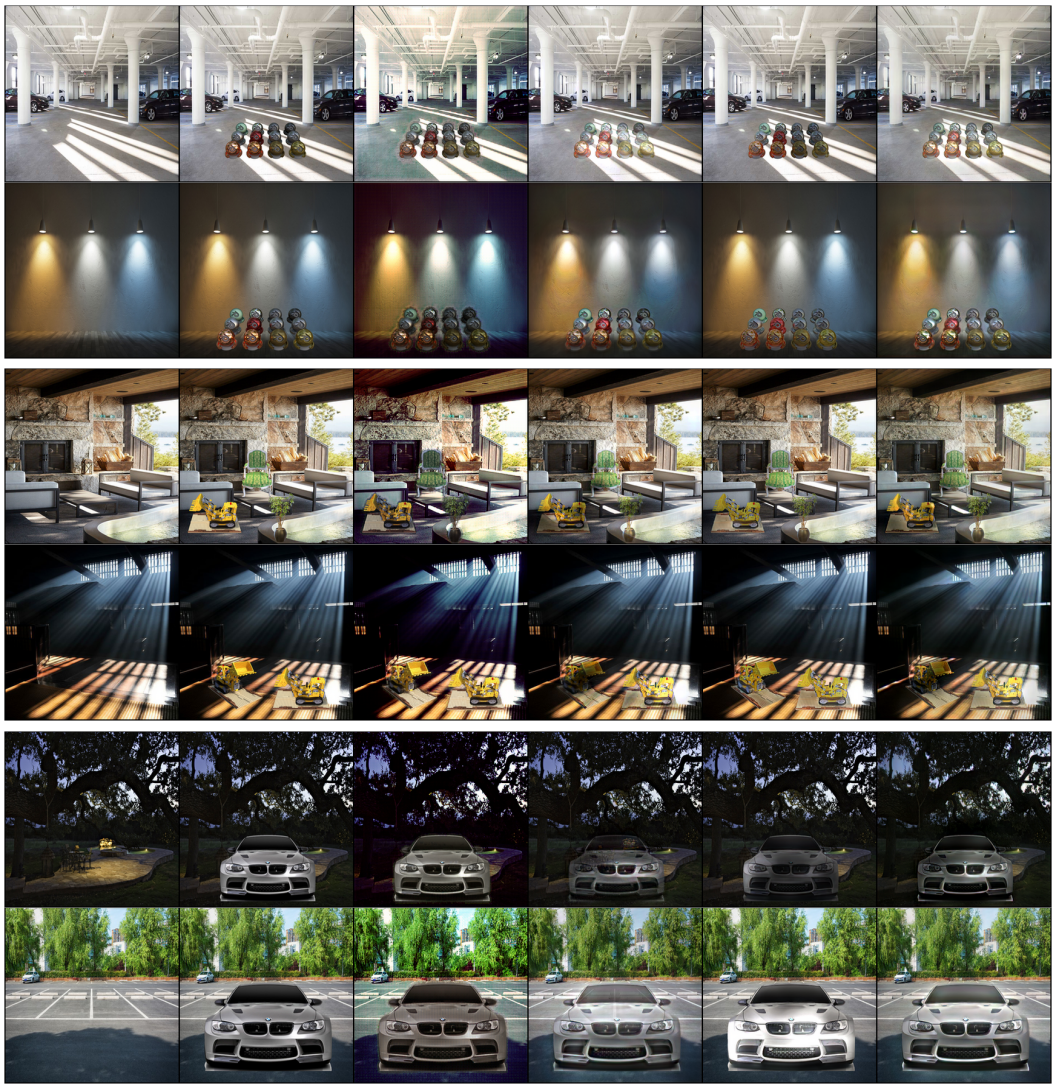


Figure 4: **Complex objects.** We show renderings of various objects – a set of 16 materials with different reflective properties in first two rows; a lego bull-dozer, a chair and a plant in the third row; two lego dozers with different poses in the fourth row and cars in the last two rows. IH optimizes for colors and results in large color shift in the background. Naive image composition works well if the shading of the source image does not clash with the rendered object. If they do, then strong shading effect the target scene appears on the object and make them look unrealistic (see the Naive Comp car in the second last row and plant in the third row). DIP suffers in scenes when there are strong light-effects (first-row) or if the scene does not have enough normals in all three directions (last-row). We trained IlluminatOR only to learn CIF without relying on external labels like normals. Hence, IlluminatOR do not suffer from these problems and convincingly reshades these complex objects. Best viewed on screen for details.

shading and gloss fields of completely arbitrary random images. This is ill-posed because the spatial appearance changes within images are strongly entangled between shading and albedo. Further, the geometry information (depth discontinuity etc) is mostly available in shading. Shading has most of the shape information of the scene and albedo has color and small local changes. Therefore, for an ideal image-relighting system with swapping light styles, the method must first learn to extract decorrelated representations and build a notion of 3D. To our surprise, the method only trained to compose image relights for a wide variety of scenes reasonably well; but also for a few cases

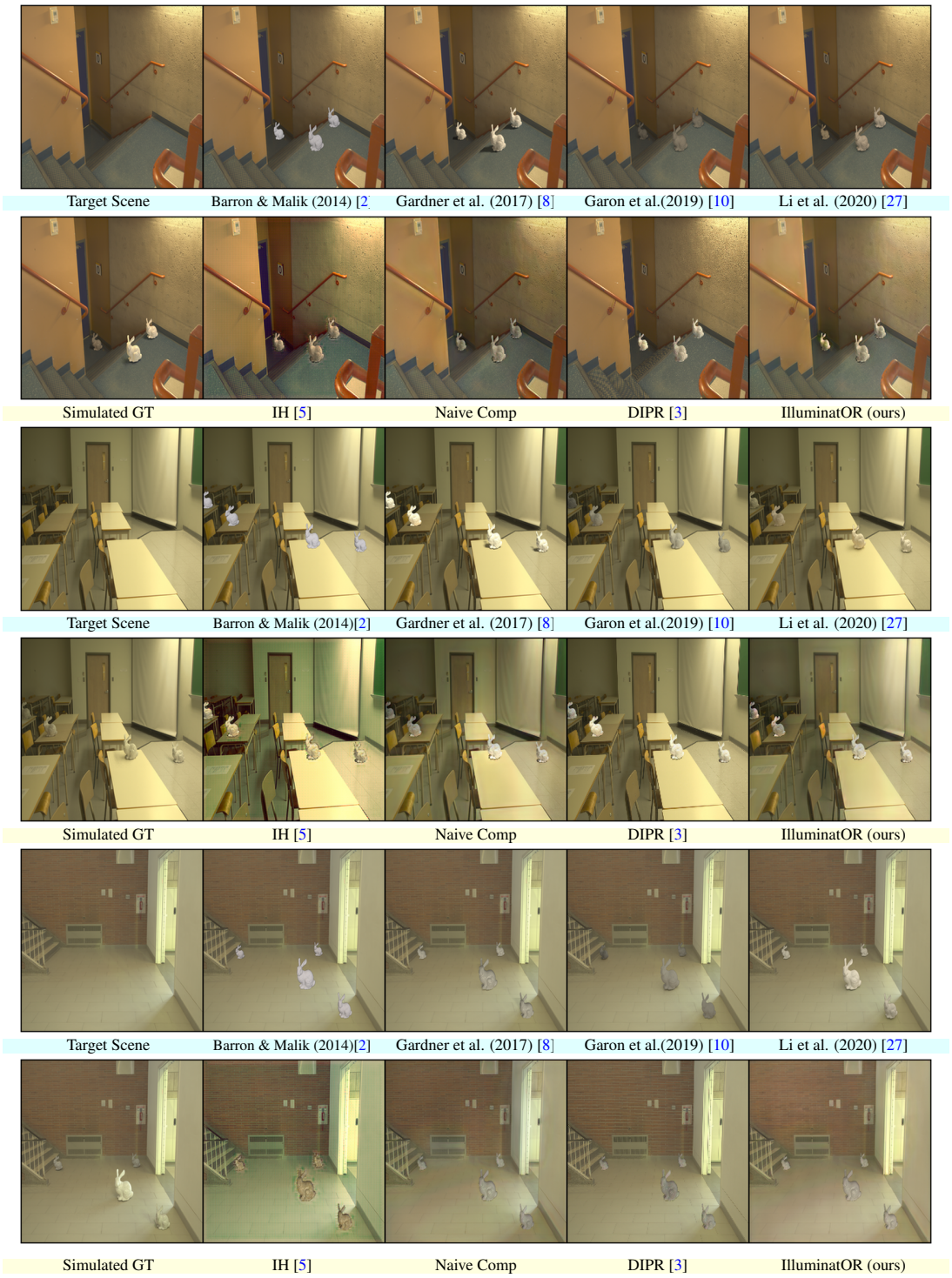


Figure 5: Qualitative comparison of image-based reshading methods with SOTA image insertion methods which require a geometric and physical model of the inserted objects (the bunnies). We use Gardner's rendering bunnies to get cut-and-paste albedo for all image-based methods. IlluminatOR produces plausible shading in each case and at times performs better than several SOTA methods. Quantitative comparisons for various albedo source choices is available in Tab 1.

Source	Sup.	IH	Naive	Ours
Barron et al. [2]	9.89	6.81	**6.69**	7.57
Gardner et al. [8]	6.83	6.68	**5.54**	5.83
Garon et al. [10]	4.39	6.67	4.77	**4.29**
Li et al. [27]	3.79	4.20	4.39	**3.93**

(a) RMSE↓

Source	Sup.	IH	Naive	Ours
Barron et al. [2]	28.76	31.72	**31.86**	30.91
Gardner et al. [8]	32.32	32.25	**33.77**	33.46
Garon et al. [10]	36.36	35.84	35.29	**36.38**
Li et al. [27]	37.22	36.37	35.80	**36.80**

(b) PSNR↑

Table 1: Quantitative Evaluation for inserting bunnies supports IlluminatOR (**Left:** RMSE; **Right:** PSNR). We extract the bunny albedo from images generated by method in the **source** column. RMSE and PSNR for the actual rendering from the respective methods in the Sup. column. These methods have various forms of supervisions (geometry, material and environment maps in training) while ours and all other baselines do not. Numbers in **bold** refers to the best performing image-based method . Numbers in underline signify that the method beats the source method when starting from them as priors. We report these numbers for pixels lying on the reshaded objects alone; the nasty shifts in background (IH) are not captured by this evaluation.

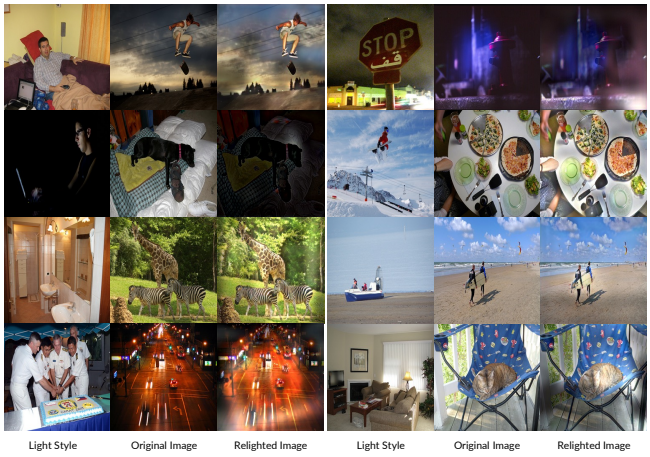


Figure 6: **Relighting with light styles.** We swap shading+gloss of the original image with another randomly sampled image from MSCOCO. IlluminatOR produces plausible relighted images. Note these images are not geometrically consistent (for example, in the second last row, for the relighted image, the change in lighting direction on beach do not adjusts the cast shadows of surfers.)

they are geometrically inconsistent. This further reinforces that our model can form a very strong compressed representation of lighting from a single 2D image. We think these CIF can further exploit representations for various "3D aware" image editing tasks. Therefore, we believe our CIF formulation is the right way to think about general-purpose image-based light editing tasks. We hope our approach to learning and injecting the scene's compressed illumination representation field from single 2D images will inspire future work for various image-based manipulations and editing.

Limitations. IlluminatOR has several strong improvements compared to DIPR and IH baselines but still trails behind methods that use geometric and physical models to render objects. For example, IlluminatOR can add small cast shadows and make subtle realistic changes in the background which DIPR cannot (see Fig. 1). However, IlluminatOR does not seem to make convincing soft shadows like physically based rendering methods. IlluminatOR clearly has some form of 3D representation, but we do not know how to extract it. Finally, evidence of generalization is purely empirical - there is no theoretical reason we are aware of that IlluminatOR behaves as it does.

Social Implications. Reshading inserted objects is a core problem in computer vision. Our system has no current user base, but we expect users to be authors of pictures. Societal consequences of the broad stream of image authoring research likely include easier production of fake images for both good and bad reasons; however, there is already interesting evidence that vast quantities of images are faked, and users know it; we do not believe that this paper changes these effects. One can use this technology in an unethical fashion, but does not itself involve ethical violations. Beneficiaries include image authors. It is possible that this work would put at a disadvantage credible consumers of images. The consequences of failure are minimal. We do not believe that the task or method leverages biases in the data, because the training regime uses so little data that such biases are likely invisible to the method.

IlluminatOR: Learning Compressed Illumination Fields for Object Reshading – Supplementary Material

Network architecture. We use a U-Net with skip connections to learn a mapping from albedos to images as shown in our Fig 2. We use 256×256 resolution images to train our model. The albedo encoder has seven convolutional blocks. The first convolutional block is without any down-sampling layer and the remaining six blocks with a down-sampling layer. Each down-sample block consists of three convolutions with three group normalization layers and a max pool layer. We use ReLU activations after each normalization layer. We extract albedo feature maps of resolution 4×4 with 512 channels from the albedo encoder.

The shading and gloss encoders are similar to albedo encoder but with only six convolutional blocks and produces features maps of 8×8 resolution with 256 channels for both the shading and the gloss fields. We concatenate both the shading and gloss convolutionally down-sampled feature maps to form a spatial "light-styles" of size $8 \times 8 \times 512$.

Our image decoder has six up-sampling convolutional blocks. Similar to down-sampling blocks, each up-sampling block consists of an up-sampling layer, three convolutional layers with group normalization and ReLU activations. The first up-sampling block directly up-samples the albedo feature map obtained. We then re-normalize this up-sampled feature using SpAdaIn with light-styles. Now before up-sampling with remaining blocks, we also concatenate albedo features of the same resolution (skip connections) which are then followed by a up-sampling block and a SpAdaIn re-normalization with light-styles. Using them we construct a feature map of size $256 \times 256 \times 3$. Finally, we apply two additional convolutions with ReLU activation to this feature map to form an image of size $256 \times 256 \times 3$.

We use ViT architecture [6] for our discriminator (D) with 16x16 unflattened image patches on a regular grid[2]. Our transformer encoder comprises of eight layers and each with eight attention heads. We use embeddings of size 256 throughout for the MLPs. Finally, we extract the embedding from the [CLS] token (256 dimension) and add a fully connected layer as head to predict a single dimension output. We apply adversarial losses to this single dimension output.

We use learning rate of 10^{-4} with an Adam optimizer.

Source	Sup.	$-\mathcal{L}_{decomp}$	$-\mathcal{L}_{adv}$	$\mathcal{L}_{comp}$
Garon et al. [10]	4.39	4.32	4.36	**4.29**
Li et al. [27]	3.79	3.99	4.05	**3.93**

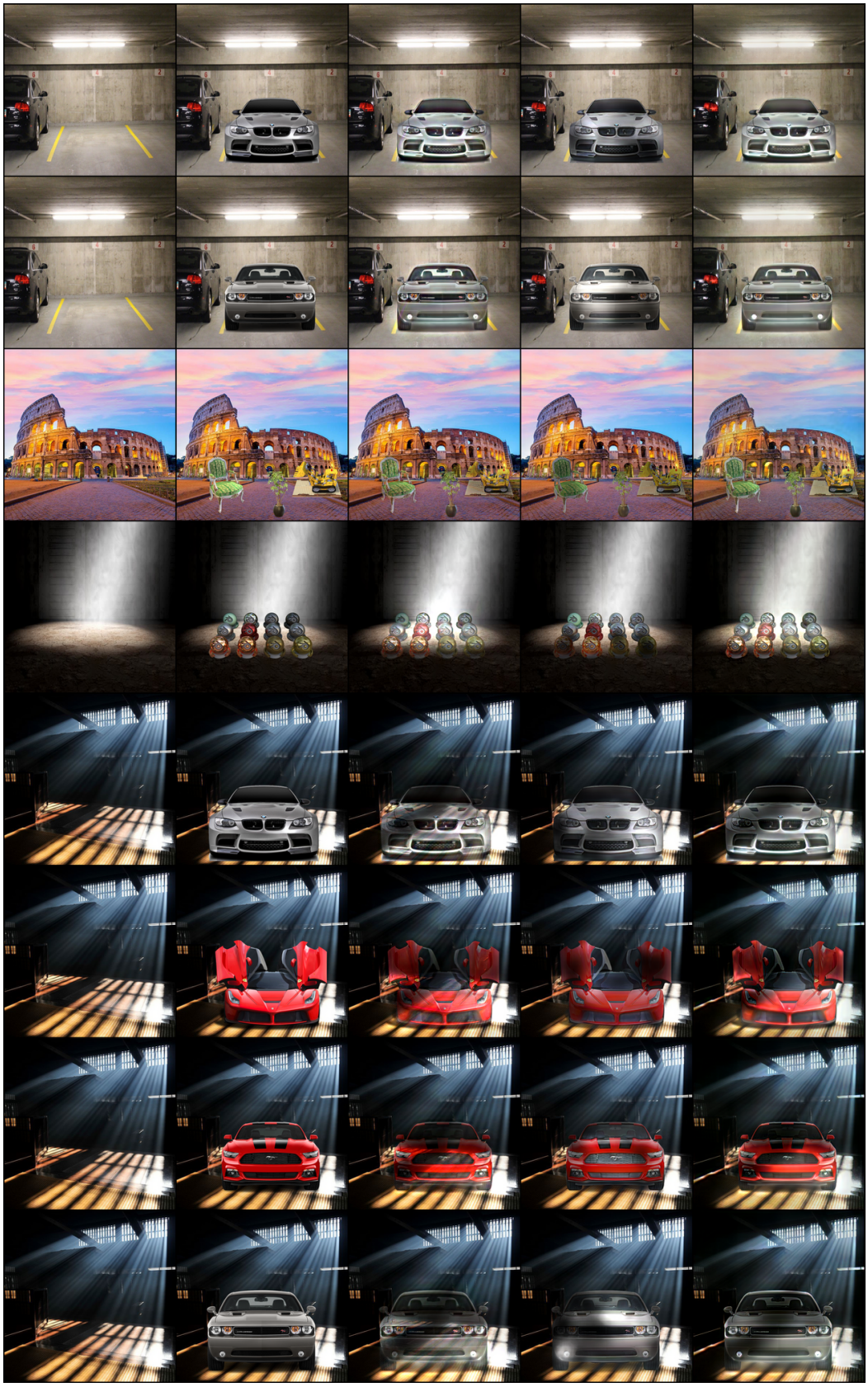
(a) RMSE↓

Source	Sup.	$-\mathcal{L}_{decomp}$	$-\mathcal{L}_{adv}$	$\mathcal{L}_{comp}$
Garon et al. [10]	36.36	36.34	36.30	**36.38**
Li et al. [27]	37.22	36.75	36.58	**36.80**

(b) PSNR↑

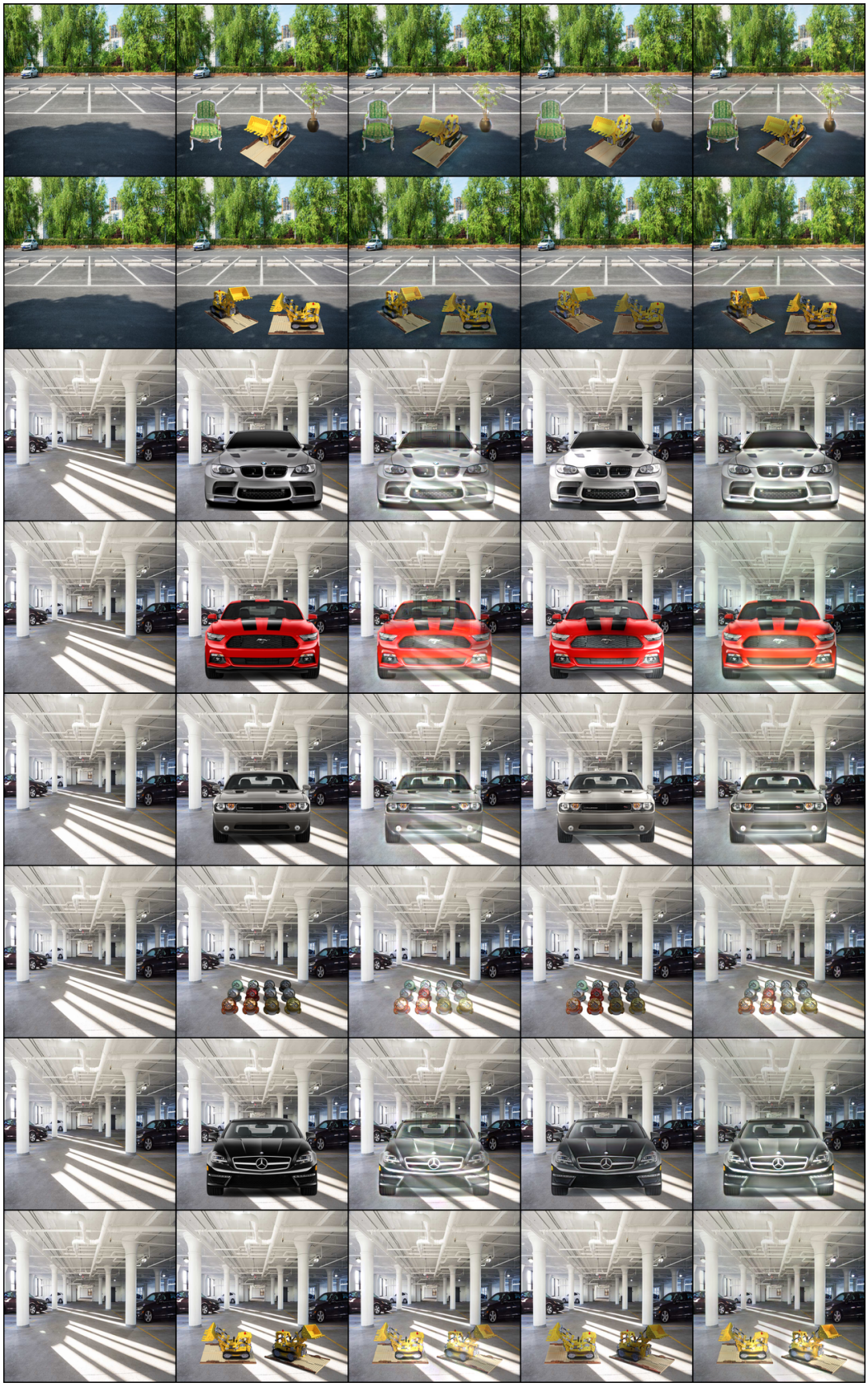
Table 2: Loss ablation study (**Left:** RMSE; **Right:** PSNR). We extract the bunny albedo from images generated by method in the **source** column. We report RMSE and PSNR for our models when trained without decomposition losses ($-\mathcal{L}_{decomp}$), without adversarial losses ($-\mathcal{L}_{adv}$) and also with all our losses ($\mathcal{L}_{comp}$). Numbers in **bold** refers to the best performing image-based method. Numbers in underline signify that the method beats the source method when starting from them as priors. Image decomposition loss helps us marginally and adversarial loss significantly improves IlluminatOR's object rendering. RMSE and PSNR for the actual rendering from the respective methods is in the Sup. column (since they are supervised with geometry and lighting parameters).

[2]https://github.com/lucidrains/vit-pytorch



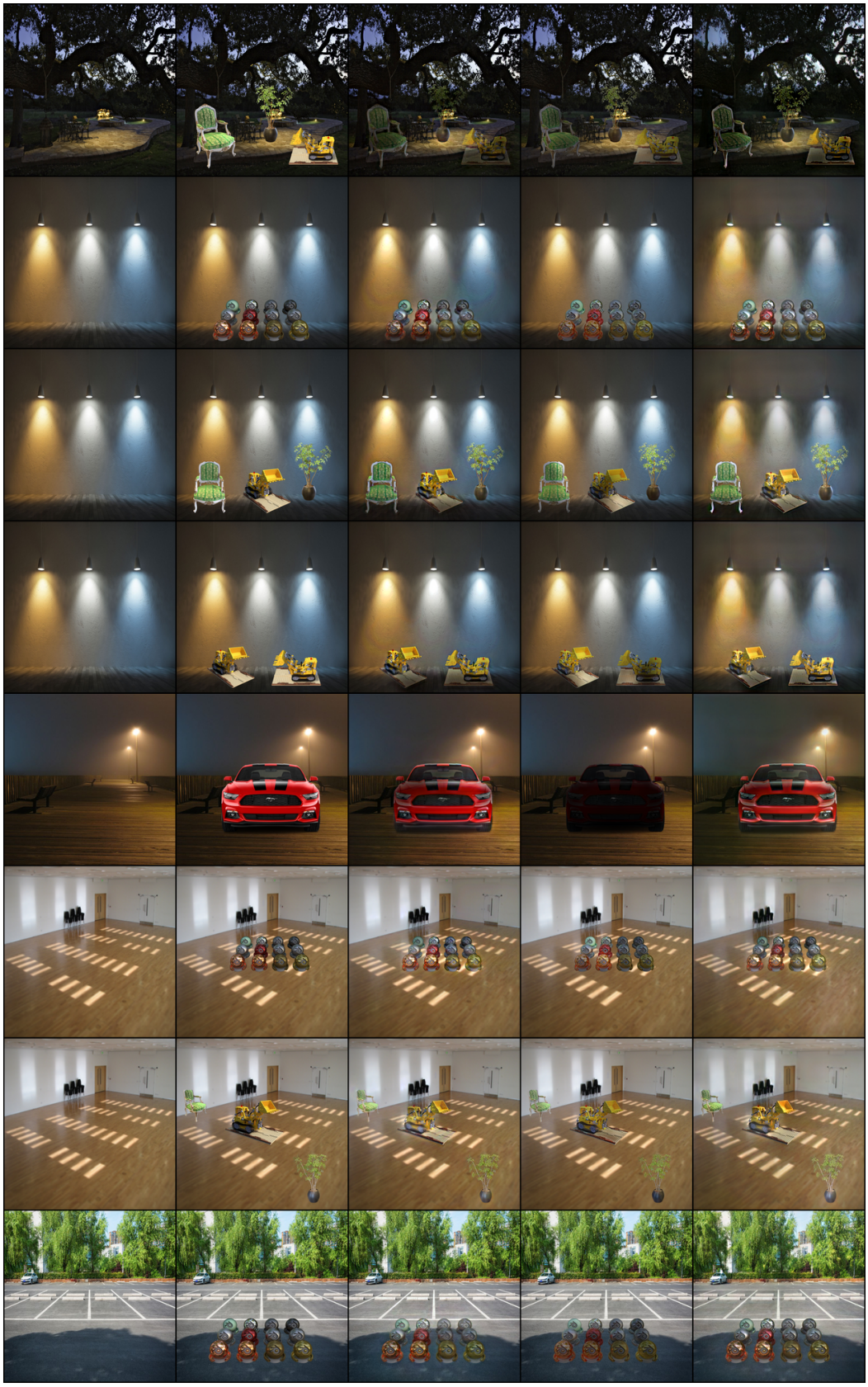
Target Scene | Cut-and-Paste | Naive Comp | DIPR [3] | IlluminatOR (ours)

Figure 7: Additional Qualitative Figures.



Target Scene | Cut-and-Paste | Naive Comp | DIPR [3] | IlluminatOR (ours)

Figure 8: Additional Qualitative Figures.



Target Scene | Cut-and-Paste | Naive Comp | DIPR [3] | IlluminatOR (ours)

Figure 9: Additional Qualitative Figures.

Target Scene | Cut-and-Paste | Naive Comp | DIPR [3] | IlluminatOR (ours)

Figure 10: Additional Qualitative Figures.

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